

OBLIGATORY TRIPLE SYSTEMS: CANONICAL ATOMS AND EXACT FINITE SPECTRA

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ABSTRACT. We classify finite obligatory triple systems and formalise the finite structural core of the classification in Lean 4. After isolated vertices are deleted, a triple system is obligatory exactly when it is linear, every hyperedge-node of its Levi graph is incident with a bridge, and every Berge cycle has even length. The associated bridge decomposition yields a canonical forest of atoms, each of which is either one triple or the private-vertex expansion of a finite 2-connected bipartite graph. This normal form determines the exact order–size–component spectrum, together with exact indecomposability, cycle-rank, and atom-count spectra. The possible atom counts form an interval, apart from a parity obstruction at cycle rank one. The negative direction is proved using a one-apex sequence lift. All arguments are in ZFC.

INTRODUCTION

Although the host systems are infinite, the classification is determined by finite structure. A finite triple system is *obligatory* if it occurs in every triple system of uncountable chromatic number. Erdős asked for a characterisation of these finite systems [1]; the question is catalogued as Erdős Problem 593 [2]. We give both constructive and intrinsic descriptions of the answer.

For a finite graph J , write J^+ for its private-vertex expansion. Let $\mathcal{B}$ be the smallest class containing J^+ for every finite bipartite graph J and every finite edgeless triple system, and closed under finite disjoint unions and one-point amalgamations. Membership in $\mathcal{B}$ is understood up to triple-system isomorphism. If F is a triple system, let F° denote the system obtained by deleting its isolated vertices.

Theorem A. *For every finite triple system F , the following are equivalent:*

- (1) F is obligatory;
- (2) $F \in \mathcal{B}$;
- (3) F° is linear, every hyperedge-node of $I(F^\circ)$ is incident with a bridge, and every Berge cycle of F° has even length.

The present proof and Lean implementation were developed independently, in parallel with closely related work by Li. Li’s preprint, posted on 23 June 2026, gives the same classification [3, Theorem 1.1] and an alternative complete-rank one-apex lift and bridge-trace formulation [3, Sections 3–4 and Theorem 4.6]. We cite this work for chronology and comparison; none of its results is used below as a black box. Our approach supplies direct positive arguments, a separate fibre-decomposition proof for the negative direction, a canonical atom normal form, exact finite structural spectra, and a Lean formalisation of the finite classification.

The graph analogue was proved by Erdős and Hajnal: the obligatory finite graphs are exactly the bipartite graphs [4, Corollary 5.6 and Theorem 7.4]. The classical nonlinearity obstruction for uniform hypergraphs is due to Erdős, Hajnal, and Rothschild [5, Theorem 2, p. 532]. Obligatory triple systems were studied further by Erdős et al. [6], Komjáth [7], Hajnal and Komjáth [8], and Komjáth [9]; the expansion theorem of Reiher [10, Theorem 1.2] supplies the positive atoms used below.

The proof separates into positive and negative directions. For the positive direction, a probabilistic rainbow lemma provides the local injectivity needed to force complete bipartite expansions. A rooted-abundance lemma then establishes closure under one-point amalgamation, including at singular uncountable cardinals. After the bridges of the Levi graph are deleted, each remaining active component

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is the expansion of a finite bipartite graph. The quotient is a forest, whose running-intersection property determines a valid amalgamation order.

For the negative direction, we use an independently developed one-apex sequence lift; Li [3, Sections 3–4] gives a parallel formulation. The lift transfers uncountable chromatic number from a graph to a triple system while controlling every finite linear trace. The trace theorem is proved by decomposing the trace into expansion fibres whose support-incidence graph is a forest, so distinct fibres can meet only at cut points. Consequently, neither a bridge-free hyperedge-node nor a Berge cycle can be assembled across fibres. Each of the three failures of the intrinsic criterion has a corresponding host: an Erdős–Rado linear host excludes nonlinearity, the lift of K_{ω_1} excludes a missing bridge, and a classical Erdős–Hajnal graph of uncountable chromatic number whose odd girth exceeds a prescribed finite bound excludes an odd Berge cycle.

All embeddings are injective and non-induced, and the argument is carried out in ZFC. The only imported results used as black boxes are the de Bruijn–Erdős compactness theorem [11, Theorem 1, pp. 371–373], the Erdős–Hajnal high-odd-girth theorem [4, Theorem 7.4, p. 76], and the pair case of the Erdős–Rado partition theorem [12, Theorem 4(i), formula (95), p. 471].

1. PRELIMINARIES

A *triple system* is a simple 3-uniform hypergraph $H = (V(H), E(H))$, so $E(H) \subseteq [V(H)]^3$. A colouring $c : V(H) \rightarrow \lambda$ is *proper* when no hyperedge is monochromatic, and $\chi(H)$ is the least cardinal λ admitting such a colouring. An embedding of a triple system F in H is an injective map $f : V(F) \rightarrow V(H)$ such that $f[e] \in E(H)$ for every $e \in E(F)$; embeddings are not required to be induced. A finite triple system F is *obligatory* if every triple system H with $\chi(H) > \aleph_0$ contains an embedding of F . A point is *isolated* if it belongs to no hyperedge, and F° denotes the result of deleting all isolated points.

All ordinary graphs in this paper are simple. A triple system is *linear* when any two distinct hyperedges meet in at most one point. Its *Levi graph* $I(F)$ is the bipartite graph with classes $V(F)$ and $E(F)$, where a point-node p is adjacent to a hyperedge-node e exactly when $p \in e$. A *bridge* is an edge of a graph whose deletion increases the number of connected components. A *Berge cycle of length* $\ell \geq 2$ consists of distinct points $p_0, \dots, p_{\ell-1}$ and distinct hyperedges $e_0, \dots, e_{\ell-1}$ such that $p_i \in e_i \cap e_{i+1}$, with indices modulo ℓ . Equivalently, it is a simple cycle of length 2ℓ in $I(F)$.

For a finite simple graph J , its *private-vertex expansion* J^+ has vertex set

$$V(J) \dot{\cup} \{p_a : a \in E(J)\}$$

and hyperedges $\{u, v, p_{\{u,v\}}\}$ for $\{u, v\} \in E(J)$, where the points p_a are new and pairwise distinct. A *one-point amalgamation* of vertex-disjoint triple systems F_0, F_1 is obtained by choosing $x_i \in V(F_i)$, identifying x_0 with x_1 , and making no other identifications or new hyperedges. Finite disjoint unions and finite edgeless systems include the empty cases.

Finite reductions.

Lemma 1.1 (Isolated-vertex reduction). *Let F be a finite triple system and let F° be obtained by deleting all isolated vertices. Then*

$$F \text{ is obligatory} \iff F^\circ \text{ is obligatory,}$$

and

$$F \in \mathcal{B} \iff F^\circ \in \mathcal{B}.$$

Proof. If F° is obligatory and H has uncountable chromatic number, then H has infinitely many vertices. After embedding the finite system F° , map the isolated vertices of F injectively to unused host vertices. Containment is non-induced, so additional host edges are irrelevant. The converse follows because F° is a subhypergraph of F .

For membership in $\mathcal{B}$, the implication $F^\circ \in \mathcal{B} \Rightarrow F \in \mathcal{B}$ follows by taking the disjoint union with a finite edgeless system.

For the converse, use structural induction on a construction of F in $\mathcal{B}$. If $F = J^+$, delete the isolated graph vertices of J to obtain J_0 ; then $(J^+)^{\circ} = J_0^+ \in \mathcal{B}$. If F is edgeless, its reduction is the empty edgeless system. Reduction commutes with disjoint union.

Suppose finally that F is a one-point amalgamation of F_0 and F_1 , identifying $x_i \in V(F_i)$. By induction, $F_0^{\circ}, F_1^{\circ} \in \mathcal{B}$. If both selected vertices are nonisolated in their factors, then F° is the one-point amalgamation of the two reductions at those vertices. If exactly one selected vertex is nonisolated, the isolated side disappears at the identified point and F° is the disjoint union of the two reductions. The same description applies when both selected vertices are isolated, because the identified point is then deleted. Hence $F^{\circ} \in \mathcal{B}$ in every case. $\square$

We may therefore suppress isolated vertices in the substantive argument and restore them at the end.

Lemma 1.2 (Finite deletion). *If $\chi(H) > \aleph_0$ and $S \subseteq V(H)$ is finite, then*

$$\chi(H - S) > \aleph_0.$$

Proof. Otherwise a countable colouring of $H - S$, together with finitely many fresh colours for S , would colour H countably. $\square$

Colouring and closure tools.

Lemma 1.3 (Two elementary colouring facts). *We use the following two facts repeatedly.*

1. *If $E(H) = E_1 \cup \dots \cup E_r$ and every $(V(H), E_i)$ is countably chromatic, then H is countably chromatic.*
2. *If a graph G admits an orientation in which every vertex has outdegree at most $d < \omega$, then $\chi(G) \leq 2d + 1$.*

Proof. For the first assertion, take the product of the r countable colourings.

For the second, every finite subgraph has at most $d|V|$ edges and therefore a vertex of degree at most $2d$. It is $(2d + 1)$ -colourable by greedy deletion. The de Bruijn–Erdős compactness theorem [11, Theorem 1] then gives a $(2d + 1)$ -colouring of all of G . $\square$

Lemma 1.4 (Closure-chain lemma). *Let κ be an uncountable cardinal, let $1 \leq r < \omega$, and let*

$$\Phi : [\kappa]^r \longrightarrow [\kappa]^{<\omega}$$

have uniformly finite values. There is an increasing continuous chain

$$\emptyset = M_0 \subseteq M_1 \subseteq \dots \subseteq M_i \subseteq \dots \quad (i < \text{cf } \kappa)$$

such that

$$\bigcup_{i < \text{cf } \kappa} M_i = \kappa, \quad |M_i| < \kappa,$$

and every M_i is Φ -closed:

$$a \in [M_i]^r \implies \Phi(a) \subseteq M_i.$$

Proof. Choose sets $A_i \subseteq \kappa$, each of cardinality below κ , whose union is κ . At a successor stage close $M_i \cup A_i$ under Φ , iterating the operation ω times. Since Φ is finitary and finite-valued, this does not increase an infinite cardinal below κ . At a limit stage take the union. A union of an increasing chain of sets closed under a finitary operation remains closed; and because the stage is below $\text{cf } \kappa$, its cardinality remains below κ . $\square$

Write

$$I_i = M_{i+1} \setminus M_i.$$

Then $(I_i)_{i < \text{cf } \kappa}$ partitions κ , and $M_i = \bigcup_{j < i} I_j$.

2. A BIPARTITE SUBGRAPH LEMMA

The following finite complete-bipartite lemma is a consequence of Erdős and Hajnal [4, Corollary 5.6, p. 72]. We give a short closure-chain proof because the construction will also be used later.

Lemma 2.1 (Uncountable chromatic number forces complete bipartite graphs). *For every positive integer n , every graph of uncountable chromatic number contains $K_{n,n}$.*

Proof. Suppose otherwise, and choose a $K_{n,n}$ -free graph G of uncountable chromatic number whose vertex cardinality κ is minimal.

For $A \in [\kappa]^n$, let

$$N(A) = \{v \in \kappa : v \text{ is adjacent to every member of } A\}.$$

Because G is $K_{n,n}$ -free, $|N(A)| < n$. Apply Lemma 1.4 to $A \mapsto N(A)$, and let M_i, I_i be the resulting chain and layers.

By minimality of κ , every induced graph $G[I_i]$ is countably chromatic. Let $G^\times$ consist of the edges joining distinct layers, and orient each such edge from the later endpoint toward the earlier endpoint. If $v \in I_i$ had n earlier neighbours, those neighbours would form an n -set $A \subseteq M_i$, and closure would give

$$v \in N(A) \subseteq M_i,$$

contrary to $v \in I_i$. Thus the orientation of $G^\times$ has outdegree at most $n - 1$. By Lemma 1.3, $G^\times$ is finitely colourable.

Colour each $G[I_i]$ with colours in ω , reusing the same palette on all layers, and combine this with a finite proper colouring of $G^\times$. The resulting countable product colouring is proper on all of G , a contradiction. $\square$

3. BIPARTITE EXPANSIONS

Rainbow bipartite submatrices.

Lemma 3.1 (Rainbow bipartite submatrices). *For all positive integers n, t , there exists $q = q(n, t)$ with the following property. Suppose the edges of $K_{q,q}$ are coloured, with an arbitrary colour set, so that at every vertex each individual colour occurs on fewer than t incident edges. Then there are n -element subsets X' and Y' of the two vertex classes such that all n^2 edges between X' and Y' have distinct colours.*

Proof. Choose independent uniformly random n -subsets X' and Y' . For every unordered pair P of distinct same-coloured edges, let I_P be the indicator that both edges of P lie in the selected $K_{n,n}$, and put

$$Z = \sum_P I_P.$$

Thus $Z = 0$ exactly when all selected edges have distinct colours.

Let m_γ be the number of edges of colour γ . The local bound gives

$$m_\gamma \leq (t - 1)q.$$

At a fixed vertex and for a fixed colour γ , fewer than t incident edges have colour γ . Summing $\binom{d_{v,\gamma}}{2} \leq (d_{v,\gamma})(t - 1)/2$ over the $2q$ vertices shows that the number of same-coloured pairs sharing a vertex is at most $(t - 1)q^2$. The total number of same-coloured pairs is at most

$$\sum_\gamma \binom{m_\gamma}{2} \leq \frac{1}{2} (\max_\gamma m_\gamma) \sum_\gamma m_\gamma \leq \frac{1}{2} (t - 1)q^3.$$

If the two edges share a vertex in the left class, their exact selection probability is

$$\frac{n}{q} \frac{(n)_2}{(q)_2},$$

the same formula holds when the shared vertex is on the right. If the two edges are disjoint, the exact probability is

$$\left(\frac{\binom{n}{2}}{\binom{q}{2}} \right)^2.$$

For $q \geq 2n$, these are at most $2n^3/q^3$ and $4n^4/q^4$, respectively. By linearity of expectation,

$$\mathbb{E}[Z] \leq \frac{2(t-1)n^3}{q} + \frac{2(t-1)n^4}{q}.$$

This is below 1 for sufficiently large q . Since Z is integer-valued and nonnegative, it cannot satisfy $Z \geq 1$ for every choice when $\mathbb{E}[Z] < 1$. Hence some choice has $Z = 0$, and the selected $K_{n,n}$ is rainbow. $\square$

The complete bipartite expansion atom.

Proposition 3.2 (The complete bipartite expansion atom). *For every positive integer n , the triple system $K_{n,n}^+$ is obligatory.*

Proof. Suppose not. Choose a $K_{n,n}^+$ -free triple system

$$H = (\kappa, E)$$

with $\chi(H) > \aleph_0$ and with κ minimal.

For a pair xy , write

$$d_H(xy) = |\{z : \{x, y, z\} \in E\}|.$$

Set $t = 3n^2 + 1$ and define a graph R on κ by

$$xy \in E(R) \iff d_H(xy) \geq t.$$

Claim 3.2.1 (the graph R is countably chromatic).

Otherwise Lemma 2.1 gives a $K_{n,n}$ in R , with core vertex set C , $|C| = 2n$. For each of its n^2 graph edges xy , choose a third vertex p_{xy} with

$$\{x, y, p_{xy}\} \in E,$$

avoiding C and all previously chosen private vertices. At every stage fewer than $2n + n^2 < t$ vertices are forbidden, while xy has at least t possible third vertices. Thus the p_{xy} can be chosen distinct and outside C , producing a copy of $K_{n,n}^+$, a contradiction. $\square$

Let

$$E_1 = \{e \in E : \text{some pair contained in } e \text{ is an edge of } R\}, \quad E_2 = E \setminus E_1.$$

A proper colouring of R is a proper colouring of (κ, E_1) , since every E_1 -edge contains an R -edge. Hence $\chi(\kappa, E_1) \leq \aleph_0$, and therefore

$$\chi(\kappa, E_2) > \aleph_0. \tag{3.1}$$

For a pair xy , define

$$\Phi(xy) = \begin{cases} \{z : \{x, y, z\} \in E\}, & d_H(xy) < t, \\ \emptyset, & d_H(xy) \geq t. \end{cases}$$

Apply Lemma 1.4 to Φ , obtaining M_i, I_i .

Every $e \in E_2$ has at least two vertices in its highest layer. Indeed, suppose the highest layer met e only in $z \in I_i$. Then the other pair $xy = e \setminus \{z\}$ lies in M_i . Since $e \in E_2$, $xy \notin R$, so $d_H(xy) < t$. Closure gives $z \in \Phi(xy) \subseteq M_i$, a contradiction.

Thus every E_2 -edge is either contained in one layer, or has exactly two vertices in its highest layer I_i and its third vertex in M_i . Let E_i^* be the E_2 -edges contained in I_i . Since $|I_i| < \kappa$ and H is $K_{n,n}^+$ -free, minimality of κ gives $\chi(I_i, E_i^*) \leq \aleph_0$. Consequently

$$E^* = \bigcup_i E_i^*$$

is countably chromatic. By (3.1), the remaining crossing-edge system

$$E^{**} = E_2 \setminus E^*$$

has uncountable chromatic number.

For each i , define a graph G_i on I_i by declaring $xy \in E(G_i)$ when there is a $\beta \in M_i$ such that $\{x, y, \beta\} \in E^{**}$. If every G_i were countably chromatic, colour each I_i properly in G_i , using the same countable palette on every layer. Every edge of E^{**} would then have differently coloured top-layer vertices, giving a countable colouring of (κ, E^{**}) , a contradiction. Hence some G_i is uncountably chromatic.

Choose $q = q(n, t)$ from Lemma 3.1. Lemma 2.1 gives a $K_{q,q}$ in G_i , with sides X, Y . For every $xy \in X \times Y$, choose

$$\beta_{xy} \in M_i$$

such that $\{x, y, \beta_{xy}\} \in E^{**}$, and regard β_{xy} as the colour of the edge xy .

This edge-colouring is locally $(t - 1)$ -bounded. For example, if for fixed x and fixed β there were t different vertices y with $\beta_{xy} = \beta$, then the pair $\{x, \beta\}$ would have codegree at least t . It would belong to R , forcing the triples $\{x, y, \beta\}$ into E_1 , contrary to their membership in $E^{**} \subseteq E_2$. The same argument applies at vertices of Y .

Lemma 3.1 gives n -sets $X' \subseteq X, Y' \subseteq Y$ for which the $\beta_{xy}, x \in X', y \in Y'$, are all distinct. They lie in M_i , whereas $X' \cup Y' \subseteq I_i$, so none is a core vertex. The triples

$$\{x, y, \beta_{xy}\}, \quad x \in X', y \in Y',$$

form $K_{n,n}^+$, the final contradiction. $\square$

Corollary 3.3 (Every finite bipartite expansion is obligatory). *If J is any finite bipartite graph, then J^+ is obligatory.*

Proof. Embed the two vertex classes of J in the two classes of some $K_{n,n}$. Then J^+ is a subhypergraph of $K_{n,n}^+$. Obligatoriness is downward closed under non-induced containment: a copy of the larger finite system contains a copy of the smaller one. Apply Proposition 3.2. $\square$

Corollary 3.3 also follows from Reiher [10, Theorem 1.2]. The argument above gives a direct proof in the notation used here.

4. CLOSURE PROPERTIES

Closure under disjoint unions and one-point amalgamations is known; see Komjáth [7, pp. 233–238] and the summary in Reiher [10, p. 1]. We include complete proofs because the subsequent argument uses the rooted-abundance construction explicitly.

Disjoint-union closure.

Lemma 4.1 (Disjoint-union closure). *The class of finite obligatory triple systems is closed under finite disjoint unions.*

Proof. Let F_1, F_2 be obligatory and let H be uncountably chromatic. First find a copy of F_1 . Delete its finite vertex set. By Lemma 1.2 the remaining triple system is still uncountably chromatic, so it contains F_2 . The two copies are disjoint. Iteration proves the finite case. $\square$

Rooted abundance and one-point amalgamation.

Definition 4.2 (Rooted abundance). Let (F, r) be a finite rooted triple system with $|V(F)| > 1$. For a triple system H and $m \geq 1$, let $R_m(F, r; H)$ be the set of vertices v for which there are m rooted copies of F , all sending r to v , whose off-root vertex sets are pairwise disjoint.

Lemma 4.3 (Rooted-abundance lemma). *If (F, r) is finite, $|V(F)| > 1$, and F is obligatory, and if $\chi(H) > \aleph_0$, then*

$$\chi(H[R_m(F, r; H)]) > \aleph_0.$$

Proof. Put

$$B = V(H) \setminus R_m(F, r; H).$$

For each $v \in B$, consider the family of off-root vertex sets of all copies of F rooted at v . Because $v \notin R_m(F, r; H)$, this family has matching number at most $m - 1$. Choose a maximal pairwise disjoint subfamily. It contains at most $m - 1$ members; let S_v be their union. Then

$$|S_v| \leq (m - 1)(|V(F)| - 1) =: D.$$

By maximality, S_v meets the off-root part of every rooted copy at v . Otherwise that off-root set would be disjoint from every member of the chosen subfamily and could be added to it.

Form a graph D_B on B by joining v to each member of $S_v \cap B$. If an edge is generated in both directions, assign it to one of its generators, and orient it away from that generator. Every vertex has outdegree at most D , so Lemma 1.3 gives a finite proper colouring of D_B .

Let A be one colour class. Then

$$S_v \cap A = \emptyset \quad (v \in A).$$

Consequently $H[A]$ contains no copy of F . Indeed, if φ were such a copy and $v = \varphi(r)$, its off-root part would have to meet S_v although both are subsets of A . Since F is obligatory, every F -free triple system is countably chromatic. Thus each of the finitely many colour classes of D_B induces a countably chromatic subsystem. Tagging a vertex by its finite D_B -colour and its countable colour inside that class gives a countable proper colouring of $H[B]$.

If $H[R_m(F, r; H)]$ were countably chromatic as well, tagging the two sides would give a countable colouring of H . This contradicts $\chi(H) > \aleph_0$. $\square$

Proposition 4.4 (One-point amalgamation closure). *The class of finite obligatory triple systems is closed under one-point amalgamations.*

Proof. Let F_1, F_2 be obligatory, with selected vertices r_1, r_2 , and let F be their one-point amalgamation. If one factor consists only of its selected vertex, the assertion is immediate. Otherwise set

$$m = |V(F_2)|.$$

By Lemma 4.3, the set

$$R = R_m(F_1, r_1; H)$$

induces an uncountably chromatic subsystem of every uncountably chromatic H . Hence $H[R]$ contains a copy of F_2 ; write φ for its embedding and let $v = \varphi(r_2)$.

At v there are m rooted copies of F_1 with pairwise disjoint off-root sets. The set $\varphi(V(F_2) \setminus \{r_2\})$ has $m - 1$ vertices, so it meets at most $m - 1$ of those off-root sets. One rooted F_1 -copy therefore avoids the F_2 -copy outside v . Their union is the required one-point amalgamation. $\square$

Corollary 3.3, Lemma 4.1, Proposition 4.4, and the trivial edgeless case prove

$$(2) \implies (1). \tag{4.1}$$

5. THE INTRINSIC DECOMPOSITION

We state the selected-incidence decomposition and its running-intersection argument in the explicit form used by the Lean development. For a distinct structural formulation, compare Li [3, Sections 4–5].

We prove $(2) \iff (3)$. Isolated vertices may be ignored by Lemma 1.1.

Forward preservation.

Proposition 5.1 (The intrinsic conditions are preserved by the generators and operations). *Every generator of $\mathcal{B}$ satisfies the intrinsic conditions, and the conditions are preserved under finite disjoint unions and one-point amalgamations.*

Proof. For a bipartite graph J , the expansion J^+ is linear. Every hyperedge-node is joined to its private point by a bridge. A Berge cycle cannot use a private point, so the Berge cycles of J^+ are precisely the ordinary graph cycles of J , with the same lengths; these are even. Edgeless systems satisfy the conditions vacuously, and disjoint union plainly preserves them.

Consider a one-point amalgamation. Hyperedges belonging to different factors intersect only in the identified point, so linearity is preserved. Its Levi graph is the vertex-sum of the two Levi graphs at a point-node. Every simple cycle in a vertex-sum lies wholly in one factor: a cycle crossing into the other factor would have to pass through the cut vertex twice. Consequently bridges in the factors remain bridges, and every Berge cycle lies in one factor and retains its parity. $\square$

Thus

$$(2) \implies (3). \tag{5.1}$$

Bridge-block reconstruction.

Proposition 5.2 (Bridge-block decomposition). *Suppose F is finite, has no isolated vertices, and satisfies the three intrinsic conditions. Then $F \in \mathcal{B}$.*

Proof. If $E(F) = \emptyset$, then the assumption that F has no isolated vertices gives $V(F) = \emptyset$, and F is an edgeless generator. Hence assume $E(F) \neq \emptyset$. Put $L = I(F)$, let $B(L)$ be its set of bridges, and let $\mathcal{C}$ be the set of connected components of $L - B(L)$.

Consider a hyperedge-node e . Since e is incident with a bridge and has degree three, it has at most two nonbridge incidences. It cannot have exactly one: a nonbridge incidence lies on a cycle, and that cycle must leave e through a second nonbridge incidence. Therefore every hyperedge-node has either no nonbridge incidences or exactly two.

The nontrivial components. Let $C \in \mathcal{C}$ contain a nonbridge incidence. Every hyperedge-node $e \in C$ has exactly two point-neighbours in C . Contract each such e to an ordinary graph edge between those two point-neighbours. This produces a finite graph J_C .

The graph J_C is simple; parallel edges would correspond to two hyperedges of F sharing two points, contrary to linearity. Every cycle in J_C gives a Berge cycle of the same length in F . Hence J_C has no odd cycle and is bipartite.

Each hyperedge-node $e \in C$ has a third point-neighbour p_e across a bridge. These third points satisfy:

1. $p_e \notin C$; otherwise a path in C from p_e to e , together with the incidence ep_e , would place that incidence on a cycle.
2. If $e \neq f$ lie in C , then $p_e \neq p_f$; otherwise a path in C from e to f , together with ep_e and p_ff , would place those bridge incidences on a cycle.

Define a map from J_C^+ to the subsystem P_C formed by these hyperedges as follows. Send every graph vertex of J_C to the corresponding point-node of C , send the private vertex indexed by the contracted edge associated with e to p_e , and send that expanded edge to the hyperedge e of F . The two observations above show that the vertex map is injective: the private points are pairwise distinct and lie outside C . Every e has exactly the two contracted endpoints and p_e as its three vertices, so this map is incidence-preserving and surjective on the vertices and hyperedges of P_C . Hence it is an isomorphism $J_C^+ \cong P_C$.

Components consisting of one hyperedge-node. If a hyperedge-node e has no nonbridge incidence, it is an isolated component after the bridges are deleted. Its single triple is the expansion of a one-edge bipartite graph. Thus every hyperedge of F belongs to a piece P_C isomorphic to J^+ for a finite bipartite graph J . The hyperedge sets of these pieces partition $E(F)$. Their vertex sets cover $V(F)$: every point is incident with a hyperedge, and it either lies in that hyperedge-node's bridge-deleted component or is the point endpoint of a bridge leaving it.

Quotient forest. Contract every component of $L - B(L)$. The original bridges become edges of a quotient multigraph T . Two distinct bridges cannot join the same pair of bridge-deleted components: paths inside those two components, together with the two bridges, would form a cycle containing both, contradicting that they are bridges. Thus T is a simple graph.

To see that T is a forest, suppose that it contained a cycle. Within each contracted component on this cycle, choose a path joining the endpoints of the two incident quotient edges. Concatenating these paths with the corresponding bridges gives a closed walk in L that traverses each of those bridges exactly once. This is impossible: deleting a bridge separates its endpoints, so every closed walk crosses the resulting cut an even number of times. Hence T is a forest.

A vertex $C \in V(T) = \mathcal{C}$ is called *active* if the bridge-deleted component C contains a hyperedge-node. For an active C , the piece P_C consists of the hyperedges represented in C , all point-nodes of C , and the point endpoints of bridge incidences leaving hyperedge-nodes in C . An inactive component contains no hyperedge-node and hence no incidence of $L - B(L)$, so it is a singleton point-node.

Every hyperedge now belongs to a bipartite expansion piece. It remains to order the pieces so that each new piece meets the assembled system in at most one point. The quotient forest records the attachment geometry needed for this running-intersection property.

Claim (5.2.1. Running intersection). *Every component of T contains an active vertex, because F has no isolated vertices. Choose an active root in each component, and order the active vertices by nondecreasing depth, with arbitrary order among equal-depth vertices. When an active component C is added, its piece P_C meets the union of the earlier pieces in at most one point.*

Proof of the claim. For a point $p \in V(F)$, let $X(p) \in \mathcal{C}$ be the component of $L - B(L)$ containing the point-node p . The active pieces containing p are precisely $P_{X(p)}$, when $X(p)$ is active, together with the pieces P_D at active neighbours D of $X(p)$ whose corresponding original bridge has point endpoint p . Thus the vertices of T indexing pieces that contain p are contained in the closed star centred at $X(p)$.

Suppose that $p \in P_C$ also lies in an earlier piece. If $X(p) = C$, every other piece containing p is indexed by a neighbour of C . A tree has only one neighbour of C at smaller depth, namely its parent, so only that piece can be earlier. If $X(p) \neq C$, then C is a neighbour of $X(p)$. The vertex $X(p)$ cannot be a child of C , because then both $X(p)$ and every other vertex in its closed star would be deeper than C . Hence $X(p)$ is the parent of C . In either case, p is the point endpoint of the unique parent edge of C . The same reasoning applies to every point shared with an earlier piece, so all such points coincide with this parent point p_C , and

$$P_C \cap \bigcup_{D \text{ earlier}} P_D \subseteq \{p_C\}.$$

For a root active component the intersection is empty. □

Starting with the root piece in each component, add the active pieces in this order. If the new intersection is empty, use a disjoint union. If it is the singleton $\{p_C\}$, use a one-point amalgamation at that point. Several earlier pieces may already contain the same inactive attachment point, but the new step still identifies only that one existing vertex.

The identity maps from the pieces into F assemble inductively to an incidence isomorphism onto the union of the pieces already added. By the running-intersection claim, the next piece and the preceding union are either disjoint or meet exactly at the point used in the amalgamation, so no unintended identification is introduced. The piece hyperedge sets are disjoint and partition $E(F)$, and their vertex sets cover $V(F)$. After all active pieces have been added, the resulting subsystem is F . Different components of T are combined by disjoint union, proving $F \in \mathcal{B}$. □

This proves

$$(3) \implies (2). \tag{5.2}$$

Combining (5.1) and (5.2),

$$(2) \iff (3). \tag{5.3}$$

Theorem 5.3 (Canonical atom normal form). *Let F be a finite triple system without isolated points satisfying the three intrinsic conditions in Theorem A. Its hyperedges admit a canonical partition into atoms with the following properties.*

- (1) *Every atom is either one triple or J^+ for a finite 2-connected simple bipartite graph J .*
- (2) *Two distinct atoms meet in at most one point. The bipartite incidence graph between atoms and points belonging to at least two atoms is a forest.*
- (3) *The atom partition, the nontrivial cores J up to graph isomorphism, and the atom–point forest are determined by F .*

Conversely, every forest assembly of these atoms by disjoint unions and one-point amalgamations satisfies the intrinsic conditions.

Proof. Let $L = I(F)$. A *cyclic block* is a maximal 2-connected subgraph of L containing a cycle. Since every hyperedge-node has degree three and is incident with a bridge, its degree inside a cyclic block is two. Suppressing the hyperedge-nodes in such a block C gives a finite graph J_C on the point-nodes of C . It is simple by linearity and 2-connected by standard block theory. Its cycles correspond to Berge cycles of the same lengths, so J_C is bipartite.

For a hyperedge-node $e \in C$, the third incidence is a bridge. Its point endpoint lies outside C , since otherwise that incidence would lie on a cycle. These third points are pairwise distinct as e varies in C ; otherwise a path inside C between two hyperedge-nodes, completed through the common third point, would put both bridge incidences on a cycle. The hyperedges represented in C therefore form exactly J_C^+ .

A hyperedge-node outside every cyclic block has no nonbridge incidence and contributes one single-triple atom. Thus the atoms partition the hyperedges. Two atoms cannot share two points, since paths inside the two atoms would produce a Levi cycle crossing two cyclic blocks. Similarly, a cycle in the atom–shared-point incidence graph expands to a closed Levi walk and hence to a Levi cycle crossing atom boundaries. The incidence graph is therefore a forest. Cyclic blocks and the block–cut forest are canonical, proving the forward statement. The converse follows because an atomic expansion is linear, has a private bridge at each hyperedge-node, and has only even Berge cycles, while a forest of one-point sums preserves these properties. $\square$

Call a connected triple system with at least one hyperedge *one-point indecomposable* if it cannot be written as a one-point amalgamation of two systems that both contain a hyperedge.

Corollary 5.4 (Minimal generators and one-point indecomposables). *Up to isomorphism, the class $\mathcal{B}$ is generated by finite edgeless systems, one triple, and J^+ for finite 2-connected simple bipartite graphs J , under finite disjoint unions and one-point amalgamations. A connected reduced obligatory system with at least one hyperedge is one-point indecomposable if and only if it is one triple or J^+ for a finite 2-connected simple bipartite graph J .*

Proof. The theorem decomposes every reduced member of $\mathcal{B}$ into the displayed atoms; isolated points are restored by an edgeless factor. If the atom forest has at least two atom-nodes, a leaf atom separates from the rest at its unique shared point. Conversely, one triple is indecomposable, and if J is 2-connected then deleting any point of J^+ cannot separate its hyperedge-nodes into two nonempty classes: a private point is a leaf, while a core point leaves J connected. Hence J^+ is indecomposable. $\square$

6. THE SEQUENCE LIFT

We use a one-apex construction specialised to ω_1 , presented through base fibres and their support-incidence graph to match the formalisation. Li [3, Sections 3–4 and Theorem 4.6] gives an alternative one-apex and bridge-trace formulation.

The one-apex construction. Let $G = (X, A)$ be an arbitrary graph, where $A = E(G)$ is also regarded as an alphabet. Let

$$T = A^{<\omega_1}$$

be the tree of all sequences of graph edges of countable ordinal length. For $s \in T$ and $x \in X$, write $x_s = (s, x)$.

Definition 6.1 (The one-apex lift). Define the triple system $\mathcal{L}(G)$ by

$$V(\mathcal{L}(G)) = T \times X.$$

For $s \in T$, an edge $a = \{x, y\} \in A$, a sequence $t \in T$ extending $s \frown a$, and $z \in X$, put

$$\{x_s, y_s, z_t\} \in E(\mathcal{L}(G)). \quad (6.1)$$

The two vertices x_s, y_s form the base pair and z_t is the apex. Since t properly extends s , these are three distinct vertices. In the resulting triple, the sequence node occurring twice uniquely identifies s ; its two graph labels identify the edge $a = \{x, y\}$, and the remaining vertex identifies t, z . Thus each hyperedge arises from unique data, up to interchanging x and y , and the system is simple and 3-uniform.

Chromatic and finite-trace properties.

Lemma 6.2 (Chromatic lower bound). *If $\chi(G) > \aleph_0$, then*

$$\chi(\mathcal{L}(G)) > \aleph_0.$$

Proof. Suppose $c : T \times X \rightarrow \omega$ is proper. For every $s \in T$, the restriction $x \mapsto c(x_s)$ is a countable colouring of G . Hence there is an edge

$$a_s = \{x_s^0, x_s^1\} \in E(G)$$

whose two vertices at node s have a common colour k_s .

Construct $s_\alpha \in A^\alpha$ for $\alpha < \omega_1$ by

$$s_0 = \emptyset, \quad s_{\alpha+1} = s_\alpha \frown a_{s_\alpha},$$

and at limit ordinals take unions.

If $\alpha < \beta$, then s_β extends $s_\alpha \frown a_{s_\alpha}$. If $k_{s_\alpha} = k_{s_\beta}$, take either endpoint z of the monochromatic edge at s_β . By (6.1),

$$\{(s_\alpha, x_{s_\alpha}^0), (s_\alpha, x_{s_\alpha}^1), (s_\beta, z)\}$$

is a monochromatic hyperedge. Therefore the colours k_{s_α} , $\alpha < \omega_1$, are pairwise distinct, impossible in ω . $\square$

The following trace theorem uses the fibre decomposition described above. Li gives a parallel bridge-trace formulation [3, Theorem 4.6].

Theorem 6.3 (Finite linear trace decomposition). *Every finite, not necessarily induced, linear subhypergraph K of $\mathcal{L}(G)$ is obtainable, by finite disjoint unions and one-point amalgamations, from systems J^+ where J is a finite subgraph of G .*

Proof. We use three local objects in the assembly. A *base node* is a sequence s at which a lift hyperedge begins. The two vertices with sequence coordinate s are its *core vertices*; the third vertex, whose coordinate properly extends s , is its *apex*. The *base fibre* K_s consists of all hyperedges of K based at s , together with their incident vertices. A base node is *active* when its fibre is nonempty. We first identify each fibre with an expansion atom and then show that the fibres form a forest of one-point overlaps.

First set aside every isolated vertex of K . Each is a one-vertex edgeless factor, hence isomorphic to J^+ for a one-vertex edgeless subgraph J of G , and may be restored by disjoint union at the end. We may therefore assume that every vertex of K lies in a hyperedge.

Every hyperedge of $\mathcal{L}(G)$ has a unique base node: exactly two of its vertices have sequence coordinate s , while the apex has a strictly longer sequence coordinate. We group the hyperedges of K by this coordinate and use the notation K_s from the dictionary.

Claim 6.3.1 (each base fibre is one expansion atom). For every active base node s , there is a finite subgraph J_s of G such that $K_s \cong J_s^+$.

Proof of Claim 6.3.1. For a fixed graph edge $a = \{x, y\} \in E(G)$, at most one hyperedge of K_s can have base pair $\{x_s, y_s\}$, since two such hyperedges would share two vertices and violate linearity. Let J_s be the finite subgraph of G whose edges are precisely the graph edges used by hyperedges of K_s , with vertex set consisting of their endpoints.

Define a map $J_s^+ \rightarrow K_s$ by sending each graph vertex $x \in V(J_s)$ to the core vertex x_s , and by sending the private vertex of $a = \{x, y\}$ to the apex of the unique hyperedge of K_s with base pair $\{x_s, y_s\}$. Distinct graph edges have distinct apices: if z_t were the apex for a and a' , the coordinate of t immediately after s would have to equal both a and a' . No apex is a core vertex, because an apex has sequence coordinate strictly extending s , whereas every core vertex has coordinate s . The map is therefore injective, and by construction it is bijective on hyperedges and preserves each incidence. Hence $K_s \cong J_s^+$. $\square$

Claim 6.3.2 (two fibres have at most one common point). For distinct active base nodes s and u ,

$$|V(K_s) \cap V(K_u)| \leq 1.$$

Proof of Claim 6.3.2. If a point z_t lies in both K_s and K_u , then both s and u are prefixes of t , so they are comparable. Suppose $s \subsetneq u$. A common point cannot be a core point of K_s ; it is an apex of K_s . The graph edge used by its hyperedge in K_s is the coordinate of u immediately following s . Hence all points in $K_s \cap K_u$ would be apices of the same base edge of K_s . There is at most one such apex. Therefore

$$|V(K_s) \cap V(K_u)| \leq 1. \tag{6.2}$$

$\square$

Let Q be the bipartite incidence graph of the base fibres. Its left class is the set of active base nodes. Its right class consists of points that lie in at least two fibres. Join a base node s to a point p exactly when $p \in V(K_s)$.

Claim 6.3.3 (the overlap graph is a forest). The graph Q is acyclic.

Proof of Claim 6.3.3. Suppose that Q has a cycle, and choose on it a base node s of minimum sequence length. Its two neighbouring base nodes on the cycle are proper descendants of s . The two intervening shared points are distinct, so they are distinct apices in K_s , arising from two distinct graph edges $a \neq a'$. Thus the two neighbouring base nodes lie in different immediate branches below s , one beginning with a and the other with a' .

Starting from one neighbour of s , every base node on the remaining part of the cycle lies in the same immediate branch below s . Indeed, consecutive base nodes share a point and are therefore comparable. For two comparable proper descendants of s , the first sequence coordinate after s is the same, so a length-two step through a shared point cannot move from one immediate branch to another. Induction along the remaining path therefore keeps every base node in the first branch, contradicting that its final node lies in the other branch. Hence Q is acyclic. $\square$

Root each component of Q at a base node, and order its base nodes by nondecreasing even distance from the root. For a nonroot base node s , let p_s be the point-node adjacent to s on the path to the root.

Claim (6.3.4. Base-fibre running intersection). *Every earlier factor that meets K_s meets it at p_s .*

Proof of Claim 6.3.4. Suppose K_u is earlier and contains a point $p \in K_s$. Then Q contains the length-two path $s - p - u$. Let the distance of s from the root be $2d$. The point-node p has distance either $2d - 1$ or $2d + 1$. In the second case u has distance $2d + 2$, so it cannot be earlier. Thus p has distance $2d - 1$, and the uniqueness of the path to the root in the tree Q gives $p = p_s$. This also covers an earlier sibling u , which shares the same parent point p_s . $\square$

Claim 6.3.5 (the ordered fibres assemble to K). Let U_s be the union of the factors preceding K_s . Equation (6.2) and Claim 6.3.4 give

$$V(K_s) \cap U_s \subseteq \{p_s\}.$$

Starting with the root factor, add the remaining factors in this order by disjoint unions or one-point amalgamations as appropriate. At each step the natural maps into K identify exactly the displayed common

point and nothing else, so they assemble to an isomorphism onto the union of the factors already added. Different components of Q , and the isolated factors removed at the start, are combined by disjoint union. This is the running-intersection step: Claim 6.3.1 supplies the expansion atoms, while Claims 6.3.2–6.3.4 certify that each new atom is attached along at most one previous point. $\square$

Corollary 6.4 (Restrictions on finite linear traces). *Every finite linear $K \subseteq \mathcal{L}(G)$ satisfies:*

1. *every hyperedge-node of $I(K)$ is incident with a bridge;*
2. *every Berge cycle of K is contained in a single factor J_s^+ ;*
3. *a Berge cycle of length ℓ in K yields an ordinary cycle of length ℓ in G .*

Proof. In every J_s^+ the incidence with the private vertex is a bridge. Bridgehood is preserved by one-point amalgamation. A simple Levi cycle cannot cross a one-point cut, so every Berge cycle lies in one factor. Berge cycles in J_s^+ are exactly ordinary graph cycles in J_s . $\square$

7. A CLASSICAL HIGH-ODD-GIRTH INPUT

For the odd-cycle obstruction we use the following classical graph theorem directly, without passing through a shift-graph surrogate.

Theorem 7.1 (Erdős–Hajnal). *For every positive integer m there is a graph G such that*

$$\chi(G) > \aleph_0$$

and G contains no odd cycle of length at most m .

Source. Theorem 7.4 of Erdős and Hajnal [4, p. 76], as quoted in Erdős et al. [6, Theorem C, p. 428], gives for every positive integer i an uncountably chromatic graph containing no cycle C_{2j+1} for $0 < j < i$. Choose i so that $2i - 1 \geq m$. $\square$

8. AVOIDANCE HOSTS

The intrinsic criterion can fail in three ways: nonlinearity, a missing bridge, or an odd Berge cycle. Li [3, Sections 2–4] organises the negative direction in a parallel way. We prove the first two avoidance constructions in our notation and apply the Erdős–Hajnal theorem from Section 7 to the third.

We prove

$$\neg(3) \implies \neg(1).$$

Nonlinearity obstruction.

Proposition 8.1 (Avoidance of every nonlinear finite triple system). *There is an uncountably chromatic linear triple system. Consequently, every finite nonlinear triple system is non-obligatory.*

Proof. We use the Erdős–Rado relation [12, Theorem 4(i), formula (95), p. 471]

$$(2^{\aleph_0})^+ \longrightarrow (\aleph_1)_{\aleph_0}^2. \quad (8.1)$$

Let $\kappa = (2^{\aleph_0})^+$. Define a triple system T_κ whose vertices are the pairs in $[\kappa]^2$, and whose hyperedges are

$$\{\{\alpha, \beta\}, \{\alpha, \gamma\}, \{\beta, \gamma\}\}, \quad \alpha < \beta < \gamma < \kappa.$$

This triple system is linear: two distinct triangles of a complete graph cannot share two graph edges, because two graph edges of a triangle determine the third edge and the three underlying vertices.

It is uncountably chromatic. A countable vertex-colouring of T_κ is a countable edge-colouring of K_κ . By (8.1) there is a monochromatic subset of cardinality $\aleph_1$, and in particular a monochromatic graph triangle. This is a monochromatic hyperedge of T_κ .

If a finite triple system F is nonlinear, it has two distinct edges sharing at least two vertices. No injective embedding of those edges can exist in the linear host T_κ . Thus T_κ is F -free. $\square$

Missing-bridge obstruction.

Proposition 8.2 (Avoidance when the bridge condition fails). *Let F be finite and linear, and suppose some hyperedge-node of $I(F^\circ)$ is incident with no bridge. Then F is non-obligatory.*

Proof. This is the K_{ω_1} specialisation of the sequence lift defined above. Li gives an alternative missing-bridge obstruction [3, Section 4]. By Lemma 1.1 it suffices to avoid F° , so assume that F has no isolated vertices. Take $G = K_{\omega_1}$. Then $\chi(G) > \aleph_0$, and Lemma 6.2 gives

$$\chi(\mathcal{L}(G)) > \aleph_0.$$

If F appeared in $\mathcal{L}(G)$, retain exactly the host hyperedges corresponding to the edges of F . Because the embedding is injective and F is linear, these selected host hyperedges form a finite linear subhypergraph isomorphic to F . Corollary 6.4 says that every hyperedge-node of such a trace is incident with a bridge, contradicting the chosen hyperedge of F . Therefore $\mathcal{L}(K_{\omega_1})$ is an uncountably chromatic F -free host. $\square$

Odd-cycle obstruction.

Proposition 8.3 (Avoidance of an odd Berge cycle). *Let F be finite and linear, and suppose F° has an odd Berge cycle. Then F is non-obligatory.*

Proof. Suppress isolated vertices and put $m = |E(F)|$. By Theorem 7.1 choose a graph G with $\chi(G) > \aleph_0$ and no odd cycle of length at most m . Lemma 6.2 gives $\chi(\mathcal{L}(G)) > \aleph_0$.

Suppose that F embeds in $\mathcal{L}(G)$ and retain exactly the host hyperedges corresponding to the edges of F . The selected trace is finite and linear. Corollary 6.4 localises every Berge cycle in one expansion fibre and sends a Berge cycle of length ℓ to an ordinary cycle of the same length in G . The given odd Berge cycle has $\ell \leq m$, contradicting the choice of G . Thus $\mathcal{L}(G)$ is an uncountably chromatic F -free host. $\square$

The three propositions cover every failure of the intrinsic conditions and therefore prove

$$\neg(3) \implies \neg(1). \tag{8.2}$$

9. PROOF OF THE CLASSIFICATION

Sections 3 and 4 establish $(2) \implies (1)$, Section 5 establishes $(2) \iff (3)$, and Section 8 proves the contrapositive $\neg(3) \implies \neg(1)$. Together these implications prove all equivalences in Theorem A. In particular,

$$F \text{ is obligatory} \iff F \in \mathcal{B},$$

$$F^\circ \text{ is linear,}$$

$$F \in \mathcal{B} \iff \begin{array}{l} \text{every hyperedge-node of } I(F^\circ) \text{ is incident with a bridge,} \\ \text{every Berge cycle of } F^\circ \text{ has even length.} \end{array}$$

All embeddings are non-induced, so additional host hyperedges on the image vertices are irrelevant. The proof uses only ZFC, graph-colouring compactness, the classical Erdős–Hajnal high-odd-girth theorem, and the standard Erdős–Rado relation (8.1). It assumes neither CH nor GCH and uses no forcing axiom or large-cardinal hypothesis.

10. FINITE PARAMETER CONSEQUENCES

In this section, a triple system is *connected* when its Levi graph is connected. We first exclude isolated vertices. For an integer $r \geq 1$, set

$$q(r) = \lceil 2\sqrt{r} \rceil.$$

Proposition 10.1 (Edge-deletion form of the bridge condition). *Let F be a finite triple system and let $e = \{x, y, z\} \in E(F)$. The hyperedge-node e has no incident bridge in $I(F)$ if and only if x, y, z lie in one connected component after the hyperedge e is deleted and all points are retained. Consequently the bridge condition in Theorem A is equivalent to the following hypergraph-native condition:*

for every $e \in E(F^\circ)$, deleting e separates its three points into at least two components.

Proof. Fix $x \in e$. The Levi incidence xe is not a bridge precisely when there is an alternative path from e to x after that incidence is deleted. A simple such path leaves e through y or z and thereafter lies entirely in the Levi graph of $F - e$. Thus xe is non-bridging exactly when x is connected in $F - e$ to at least one of the other two points of e .

If all three incidences at e are non-bridging, no one of x, y, z can form a singleton component among those three points. A partition of three points with this property has one block, so all three are connected in $F - e$. The converse follows by reversing the same paths. $\square$

Lemma 10.2 (Bipartite shadow). *Let F be an obligatory finite triple system with no isolated vertices and at least one hyperedge. There is a finite bipartite graph J , without isolated vertices, such that*

$$|E(J)| = |E(F)|, \quad |V(J)| = |V(F)| - |E(F)|,$$

and J has the same number of connected components as F . Conversely, for every finite bipartite graph J without isolated vertices, the expansion J^+ is obligatory and has these parameter relations.

Proof. By Theorem A and Proposition 5.2, each connected component H of F is obtained by one-point amalgamations from connected expansion pieces

$$J_1^+, \dots, J_k^+,$$

where each J_i is a connected bipartite graph. Disconnected graph pieces may simply be split into their connected components. Write $m_i = |E(J_i)|$ and $s_i = |V(J_i)|$. Since H is connected, its k pieces are joined by exactly $k - 1$ one-point identifications. Hence

$$|E(H)| = \sum_{i=1}^k m_i, \quad |V(H)| = \sum_{i=1}^k (m_i + s_i) - (k - 1). \quad (10.1)$$

Take one-point sums of the ordinary graphs $J_1, \dots, J_k$ along any tree on the k indices. Before each identification, interchange the two bipartition classes of the new factor if necessary. The resulting graph J_H is connected and bipartite, and (10.1) gives

$$|E(J_H)| = |E(H)|, \quad |V(J_H)| = |V(H)| - |E(H)|.$$

Taking the disjoint union of these shadows over the components of F proves the forward assertion. The converse is the expansion-generator case of Theorem A, with the displayed counts immediate from the definition of J^+ . $\square$

Theorem 10.3 (Exact order–size–component spectrum). *Let $m \geq 1$, $1 \leq c \leq m$, and n be integers. There exists an obligatory triple system F with no isolated vertices, exactly m hyperedges, exactly n vertices, and exactly c connected components if and only if*

$$\boxed{m + 2(c - 1) + \left\lceil 2\sqrt{m - c + 1} \right\rceil \leq n \leq 2m + c.} \quad (10.2)$$

Every integer in this interval occurs.

Proof. We begin with the corresponding elementary graph fact. A connected simple bipartite graph with $r \geq 1$ edges and s vertices exists if and only if

$$q(r) \leq s \leq r + 1. \quad (10.3)$$

Indeed, connectedness gives $r \geq s - 1$. If the bipartition has sizes a, b , then

$$r \leq ab \leq \left\lfloor \frac{s^2}{4} \right\rfloor,$$

which is equivalent to $s \geq q(r)$. Conversely, $K_{\lfloor s/2 \rfloor, \lceil s/2 \rceil}$ contains a spanning tree, and starting with that tree and adding arbitrary unused edges realises every edge count between $s - 1$ and $\lfloor s^2/4 \rfloor$.

Apply Lemma 10.2. Let the c connected components of the shadow have positive edge counts $m_1, \dots, m_c$, where $\sum_i m_i = m$. If their orders are s_i , then (10.3) gives

$$\sum_{i=1}^c q(m_i) \leq \sum_{i=1}^c s_i \leq m + c. \quad (10.4)$$

For positive integers a, b ,

$$q(a) + q(b) \geq q(a + b - 1) + 2. \quad (10.5)$$

To see this, note that

$$\sqrt{a} + \sqrt{b} \geq \sqrt{a + b - 1} + 1,$$

because, after squaring, the assertion reduces to $(a - 1)(b - 1) \geq 0$; taking ceilings gives (10.5). Repeatedly applying (10.5) to the left side of (10.4) yields

$$\sum_{i=1}^c q(m_i) \geq q(m - c + 1) + 2(c - 1).$$

The shadow has $n - m$ vertices, so these inequalities give (10.2).

For sharpness put $M = m - c + 1$. Take $c - 1$ components equal to K_2 . By (10.3), a final connected bipartite component with M edges can have every order from $q(M)$ through $M + 1$. Their disjoint union therefore realises every shadow order from $q(M) + 2(c - 1)$ through $m + c$. Expanding it adds exactly m private vertices and proves attainability of every integer in (10.2). $\square$

Corollary 10.4 (Connected and unrestricted spectra). *For $m \geq 1$:*

- (1) *a connected obligatory triple system without isolated vertices has m hyperedges and n vertices if and only if*

$$m + \lceil 2\sqrt{m} \rceil \leq n \leq 2m + 1;$$

- (2) *without prescribing the number of components, a reduced obligatory triple system has m hyperedges and n vertices if and only if*

$$m + \lceil 2\sqrt{m} \rceil \leq n \leq 3m;$$

- (3) *if isolated vertices are permitted, an obligatory triple system with m hyperedges and n vertices exists if and only if*

$$n \geq m + \lceil 2\sqrt{m} \rceil.$$

Proof. The first assertion is Theorem 10.3 with $c = 1$. For the second, the connected interval covers through $2m + 1$; for $n > 2m + 1$, take $c = n - 2m$ and use the upper endpoint $n = 2m + c$ of (10.2). The largest possible value is obtained from m disjoint triples. The final assertion follows by adjoining arbitrary isolated vertices. $\square$

Corollary 10.5 (Size spectrum at fixed order). *Let $c \geq 1$ and $n \geq 3c$. A reduced obligatory triple system with exactly n vertices, c connected components, and m hyperedges exists if and only if*

$$\left\lceil \frac{n - c}{2} \right\rceil \leq m \leq n - 2c + 4 - \left\lceil 2\sqrt{n - 3c + 4} \right\rceil. \quad (10.6)$$

Consequently, a reduced obligatory triple system with exactly c connected components and n vertices exists if and only if

$$n = 3c, \quad n = 3c + 2, \quad \text{or} \quad n \geq 3c + 4. \quad (10.7)$$

Proof. The upper-order inequality in (10.2) is equivalent to $m \geq \lceil (n - c)/2 \rceil$. For the other inequality put

$$M = m - c + 1, \quad N = n - 3c + 3.$$

It becomes

$$M + \lceil 2\sqrt{M} \rceil \leq N. \quad (10.8)$$

For integers $N \geq 3$ and $M \geq 1$, the largest M satisfying (10.8) is

$$U = N + 2 - \lceil 2\sqrt{N+1} \rceil. \quad (10.9)$$

Indeed, write $k = \lceil 2\sqrt{N+1} \rceil$. Since $k^2 \geq 4(N+1)$,

$$4U = 4N + 8 - 4k \leq (k-2)^2,$$

so $\lceil 2\sqrt{U} \rceil \leq k-2$ and $U + \lceil 2\sqrt{U} \rceil \leq N$. On the other hand, $(k-1)^2 < 4(N+1)$ gives

$$(k-3)^2 < 4(N+3-k) = 4(U+1),$$

so $\lceil 2\sqrt{U+1} \rceil \geq k-2$ and $(U+1) + \lceil 2\sqrt{U+1} \rceil \geq N+1$. The left side of (10.8) is strictly increasing in M , proving (10.9). Translating back to m gives (10.6).

For (10.7), Corollary 10.4 shows that a connected reduced obligatory system has order 3, order 5, or any order at least 7. The first two statements come from $m = 1, 2$. For $m \geq 3$, the connected intervals have no gaps because $\lceil 2\sqrt{m+1} \rceil \leq m+1$ for $m+1 \geq 4$. Thus every component has order

$$3 + r, \quad r \in \{0, 2\} \cup [4, \infty).$$

The sum of c such increments is again 0, 2, or any integer at least 4: use one nonzero increment and take all remaining increments to be zero. The increments 1 and 3 are impossible. This proves (10.7). $\square$

Corollary 10.6 (Exact Levi cycle-rank spectrum). *Let F be reduced and obligatory, with m hyperedges, n vertices, and c connected components. The cyclomatic number of its Levi graph is*

$$\beta(I(F)) = |E(I(F))| - |V(I(F))| + c = 2m - n + c. \quad (10.10)$$

For fixed m, c , every integer in the interval

$$0 \leq \beta(I(F)) \leq m - c + 2 - \lceil 2\sqrt{m - c + 1} \rceil \quad (10.11)$$

occurs, and no other value occurs. In particular,

$$|V(F)| = 2|E(F)| + c \iff I(F) \text{ is a forest.}$$

Proof. The Levi graph has $n + m$ vertices, $3m$ incidence edges, and c components, which gives (10.10). Substitution of the two endpoints in (10.2) gives (10.11), and every intermediate value occurs because every intermediate order does. A finite graph has cyclomatic number zero exactly when it is a forest. $\square$

Corollary 10.7 (Rigidity at balanced lower endpoints). *Let F be connected, reduced, and obligatory.*

(1) *If $|E(F)| = t^2$ and $|V(F)| = t^2 + 2t$, then $F \cong K_{t,t}^+$.*

(2) *If $|E(F)| = t(t+1)$ and $|V(F)| = t(t+1) + 2t + 1$, then $F \cong K_{t,t+1}^+$.*

Here $t \geq 1$, and the complete bipartite graphs are unique up to interchanging the two classes.

Proof. In either case the shadow from Lemma 10.2 attains $|E(J)| = \lfloor |V(J)|^2/4 \rfloor$. Equality forces every possible cross edge to be present and the two bipartition classes to be balanced, so the shadow is $K_{t,t}$ or $K_{t,t+1}$, respectively.

For $t \geq 2$ these complete bipartite graphs have no cut vertex. On the other hand, the shadow built from two or more connected expansion pieces is a nontrivial one-point sum and has a cut vertex. Hence the bridge-block reconstruction has only one piece, and F is the expansion of the displayed complete bipartite graph. The cases $t = 1$ are immediate. $\square$

Canonical atoms and structural spectra. For a connected reduced obligatory system F , put

$$s = |V(F)| - |E(F)|, \quad \beta = 2|E(F)| - |V(F)| + 1.$$

Thus s is the order of a connected bipartite shadow and β is its cycle rank. Let $a(F)$ denote the number of canonical atoms from Theorem 5.3.

Lemma 10.8 (Minimum order at fixed cyclic rank). *Let J be a finite 2-connected simple bipartite graph of cycle rank $r = |E(J)| - |V(J)| + 1 \geq 1$. Then*

$$|V(J)| \geq 2 + q(r). \quad (10.12)$$

For $r \geq 2$, every order $v \geq 2 + q(r)$ occurs. For $r = 1$, the possible orders are exactly the even integers $v \geq 4$.

Proof. Writing $v = |V(J)|$, the bipartite extremal bound gives

$$r \leq \left\lfloor \frac{v^2}{4} \right\rfloor - v + 1 = \left\lfloor \frac{(v-2)^2}{4} \right\rfloor,$$

which is equivalent to (10.12). If v is even, start with the spanning cycle C_v in the balanced complete bipartite graph and add edges until the desired rank is reached. If v is odd, start with C_{v-1} and add the remaining vertex with two neighbours in the opposite bipartition class; this is 2-connected and has rank two. Add arbitrary missing cross-edges. These constructions realise every allowed $r \geq 2$. A 2-connected graph of rank one is a cycle, which is bipartite exactly when its order is even. $\square$

Theorem 10.9 (Exact canonical atom-count spectrum). *Let F range over connected reduced obligatory systems with fixed shadow order s and cycle rank β . The possible values $k = a(F)$ are exactly*

$$\begin{aligned} \beta = 0 : & \quad k = s - 1, \\ \beta = 1 : & \quad 1 \leq k \leq s - 3, \quad k \equiv s + 1 \pmod{2}, \\ \beta \geq 2 : & \quad 1 \leq k \leq s - 1 - q(\beta). \end{aligned} \quad (10.13)$$

For $\beta \geq 1$, the maximum is

$$a_{\max}(s, \beta) = s - 1 - q(\beta). \quad (10.14)$$

Every maximiser has one minimum-order cyclic atom carrying the full rank β , and every other atom is one triple. The parity obstruction in the rank-one line is the only gap in any atom-count spectrum.

Proof. Let the atom cores be J_i , with $v_i = |V(J_i)|$, $e_i = |E(J_i)|$, and $r_i = e_i - v_i + 1$. Since a tree of k atoms makes $k - 1$ one-point identifications,

$$s = 1 + \sum_i (v_i - 1), \quad \beta = \sum_i r_i. \quad (10.15)$$

If $\beta = 0$, every atom is one triple and (10.15) gives $k = s - 1$. If $\beta = 1$, there is one cyclic atom, whose core is an even cycle of order $v \geq 4$; the remaining atoms are single triples, so $s = k + v - 1$. This is exactly the middle line of (10.13).

Suppose $\beta \geq 2$. For positive integers a, b , the inequality $\sqrt{a} + \sqrt{b} \geq \sqrt{a+b-1} + 1$ follows after squaring from $(a-1)(b-1) \geq 0$. Hence $q(a) + q(b) \geq q(a+b-1) + 2 \geq q(a+b) + 1$, where the second inequality uses that q increases by at most one between consecutive integers. Together with Lemma 10.8, repeated application gives

$$s \geq 1 + k + q(\beta),$$

so $k \leq s - 1 - q(\beta)$. Conversely, for every integer in the displayed interval choose one cyclic core of rank β and order $s - k + 1$, supplied by the lemma, and attach $k - 1$ single-triple atoms. Equality in (10.14) forces one positive-rank atom and minimum order, proving the rigidity statement. $\square$

For $s \geq 3$, define

$$\alpha(s) = \begin{cases} s, & s \text{ even,} \\ s + 1, & s \text{ odd,} \end{cases} \quad \delta(s) = \left\lfloor \frac{(s-1)^2}{4} \right\rfloor + 1.$$

Theorem 10.10 (Indecomposability phase diagram). *Apart from the single triple $(|E(F)|, |V(F)|) = (1, 3)$, a connected reduced one-point-indecomposable obligatory system with shadow order s and m hyperedges exists if and only if*

$$s \geq 4, \quad \alpha(s) \leq m \leq \left\lfloor \frac{s^2}{4} \right\rfloor. \quad (10.16)$$

A connected reduced one-point-decomposable obligatory system exists if and only if

$$s \geq 3, \quad s - 1 \leq m \leq \delta(s). \quad (10.17)$$

Consequently the feasible connected parameters split into three exact zones:

$$\begin{aligned} s - 1 \leq m < \alpha(s) &\implies \text{every system is decomposable,} \\ \alpha(s) \leq m \leq \delta(s) &\implies \text{both behaviours occur,} \\ \delta(s) < m \leq \lfloor s^2/4 \rfloor &\implies \text{every system is indecomposable.} \end{aligned} \quad (10.18)$$

For $m \geq 2$, the last zone is equivalently

$$s \leq \left\lceil 2\sqrt{m-1} \right\rceil. \quad (10.19)$$

Proof. By Corollary 5.4, the nontrivial indecomposable systems are J^+ with J finite, simple, bipartite and 2-connected. If $|V(J)| = s$ and $|E(J)| = m$, minimum degree two gives $m \geq s$; equality makes J a cycle and therefore forces s even. The upper bound is the usual $m \leq \lfloor s^2/4 \rfloor$. Even cycles, and for odd s a cycle on $s - 1$ vertices plus one new vertex with two neighbours, give the lower endpoints; adding cross-edges fills the intervals.

A decomposable system admits a connected bipartite shadow with a cut vertex. If its bipartition sizes are $a + b = s$ and the cut vertex lies in the a -side, then deleting it gives components with sizes (a_i, b_i) and

$$m \leq b + \sum_i a_i b_i \leq b + (a - 1)(b - 1) = ab - a + 1.$$

The symmetric estimate gives $m \leq ab - \min(a, b) + 1 \leq \delta(s)$. The bound is sharp: take a connected bipartite graph on $s - 1$ vertices with $m - 1$ edges and adjoin one leaf. This also fills every value in (10.17). The three zones and the elementary inversion (10.19) follow. $\square$

Corollary 10.11 (Rigidity on the structural boundaries). *Let F be connected, reduced and obligatory, with shadow order s .*

- (1) *If F is indecomposable, s is even and $m = s$, then $F \cong C_s^+$.*
- (2) *If F is indecomposable, s is odd and $m = s + 1$, then $F \cong \Theta(2a, 2b, 2c)^+$ for positive $a \leq b \leq c$ with $a + b + c = (s + 1)/2$.*
- (3) *If F is indecomposable and $m = \lfloor s^2/4 \rfloor$, then $F \cong K_{\lfloor s/2 \rfloor, \lceil s/2 \rceil}^+$.*
- (4) *If F is decomposable and $m = \delta(s)$, then F is a one-point amalgamation of $K_{a,b}^+$ and one triple, where $a + b = s - 1$ and $|a - b| \leq 1$.*

Here $\Theta(p, q, r)$ is the theta graph with three internally disjoint paths of lengths p, q, r and common endpoints.

Proof. The first and third assertions are the equality cases of minimum degree and the bipartite extremal bound. In the second case $\sum_v (\deg v - 2) = 2$. A single degree-four vertex would be a cut vertex, so there are exactly two degree-three vertices and all others have degree two; the three resulting paths have the same parity, hence are all even. Equality in the cut-vertex estimate used above forces a balanced complete bipartite graph on $s - 1$ vertices plus one leaf, which translates to the fourth assertion. $\square$

Proposition 10.12 (Componentwise atom-count spectrum). *Let F be reduced and obligatory with at least one hyperedge and $c \geq 1$ connected components, shadow order $s = |V(F)| - |E(F)|$, cycle rank*

$\beta = 2|E(F)| - |V(F)| + c$, and k canonical atoms. For every feasible (s, β, c) , the possible values of k are exactly

$$\begin{aligned} \beta = 0 : & \quad k = s - c, \\ \beta = 1 : & \quad c \leq k \leq s - c - 2, \quad k \equiv s - c \pmod{2}, \\ \beta \geq 2 : & \quad c \leq k \leq s - c - q(\beta). \end{aligned} \tag{10.20}$$

The parameters themselves are feasible exactly when $s \geq 2c$ for $\beta = 0$ and $s \geq 2c + q(\beta)$ for $\beta \geq 1$.

Proof. If the atom cores are J_i , a forest with k atom-nodes and c components makes $k - c$ identifications. Hence

$$s = c + \sum_i (|V(J_i)| - 1), \quad \beta = \sum_i (|E(J_i)| - |V(J_i)| + 1).$$

The connected proof applies componentwise. Necessity gives the lower atom floor $k \geq c$ and the three upper formulas. For sufficiency, take $c - 1$ single-triple components and realise the corresponding connected atom count in the remaining component. The feasibility thresholds follow by concentrating all positive rank in one minimum-order cyclic atom. $\square$

FORMAL VERIFICATION AND REPRODUCIBILITY

The self-contained Lean 4 source is available in the public `Erdos593` repository. The public commit history records a checked formalisation scaffold dated 15 July 2026 and, later that day, the complete finite structural classification in commit `6fd00a7`. To the author’s knowledge, this is the first publicly timestamped Lean formalisation of the finite structural theorem. Li’s arXiv v2, submitted on 23 July 2026, contains a separate formal verification of the broader results in his paper. No assertion of shared code or derivation between the two Lean developments is made. For every finite triple system F , the exported theorems establish

$$\begin{aligned} \text{F.IsObligatory} &\iff \text{F.isolatedReduction.Intrinsic}, \\ \text{F.IsObligatory} &\iff \text{Constructible F.isolatedReduction}. \end{aligned}$$

These exported theorems verify both directions of the classification, including the isolated-vertex reduction. Here “finite” describes the system F being classified. The host triple systems are not assumed finite; they are quantified within the documented ambient-universe convention of the formalisation. The formalisation supplements rather than replaces mathematical review. Its exported endpoint covers Theorem A and the isolated-vertex reduction. The manuscript proves the canonical-atom normal form and the exact finite spectra, which are also supported by finite certificates, but does not claim them as additional Lean endpoints.

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